

RESEARCH INTERESTS	Soil–vegetation–atmosphere interaction; water and carbon fluxes; land surface modelling and deep learning.	
EDUCATION	PhD, Geo-Information Science and Earth Observation, University of Twente Feb 2023—... – WUNDER. Process-based land surface modelling and deep learning for water and carbon fluxes. – Supervisors: Prof. dr. Bob Su and dr. Yijian Zeng . M.Sc., Environmental Engineering, Technical University of Munich Nov 2020—Dec 2022 – Graduated 1.4/5.0. Thesis: soil moisture controls on vegetation, in <i>Biogeosciences</i> . B.E., Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal Nov 2014—Aug 2018 – Ranked 20/192, 77.7%. Thesis: flood inundation mapping, Babai River.	
EMPLOYMENT	Research Intern, Max Planck Institute for Biogeochemistry, Jena Nov 2020—Feb 2022 – Near-surface soil moisture vs. terrestrial water storage, controls on global vegetation. Student Research Assistant, Technical University of Munich Nov 2021—Mar 2022 – Flood inundation maps for the Ouémé (Benin) and Naryn (Kyrgyzstan) rivers, HEC-RAS 1D. Working Student, Jena-Geos Ingenieurbüro GmbH, Munich May 2021—Dec 2021 – Excel/VBA tool for greenhouse-gas accounting: combustion, refrigeration, transport. Civil Engineer (freelance), ECEP, Kathmandu, Nepal Dec 2018—Dec 2022 – Business plans for community water supply projects, Government of Nepal. – Smart water supply master plan for Lumbini, Ministry of Urban Development.	
PUBLICATIONS	<i>Peer-reviewed</i> Moutahir, H., Sulzer, M., Kiese, R., ... Khanal, P. , et al. (2026). Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany. <i>Biogeosciences</i> , 23, 1719–1738. Zeng, Y., Alidoost, F., Schilperoort, B., ... Khanal, P. , et al. (2025). Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science. <i>Computers & Geosciences</i> , 205, 106013. Khanal, P. , Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning. <i>Biogeosciences</i> , 21(6), 1533–1547. Bhattarai, P., Khanal, P. , Tiwari, P., et al. (2019). Flood inundation mapping of Babai basin using HEC-RAS and GIS. <i>Journal of the Institute of Engineering</i> , 15(2), 32–44. <i>Under review</i> Khanal, P. , Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model. Preprint on SSRN. Khanal, P. , et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings. <i>Conference contributions</i> Khanal, P. , et al. (2026). Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning. <i>Workshop on Machine Learning for Land and Hydrology</i> , ECMWF, Reading, UK. Khanal, P. , Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning. <i>EGU General Assembly</i> , Vienna, EGU23-7670. Khanal, P. (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. <i>International Conference on Health Engineering in Disaster</i> .	
AWARDS	Second place, Hackathon on Machine Learning for Earth System Modelling Aug 2026 – Quantization in foundation models. CESOC, ECMWF, TRA Modelling, FZ Jülich, Cologne.	
SCHOLARSHIPS	Deutschlandstipendium, national merit scholarship, TU Munich 2021/22 Merit-based scholarship, Institute of Engineering, Tribhuvan University 2014—2018	
TEACHING	Teaching Assistant, Technical University of Munich 2021—2022 – River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow. Assistant Lecturer, Cosmos College of Management and Technology, Nepal Nov 2018—Jan 2021 – Elective: Python and GIS for water resources.	
SOFTWARE	WUNDER data viewer, ITC, University of Twente 2026 – Weather and soil moisture loggers, read by growers at the food forests as drought develops. Source .	
SKILLS	<i>Modelling</i> : STEMMUS-SCOPE, land surface models, HEC-RAS 1D <i>Data</i> : eddy covariance, satellite soil moisture, ICOS <i>Programming</i> : Python, deep learning, GIS, Excel/VBA <i>Languages</i> : English (C2), German (A1), Dutch (A2), Hindi, Nepali	